ADMITERE MATE 2025

1. $2^{x+1} + 2^x = 12$

$2^x(2+1) = 12$

$3 \cdot 2^x = 12 \mid :3$

$2^x = 4 \Rightarrow x = 2$

a). $x = 2$

2. $n = ?$ a.î. $C_n^2 = 6$

$$\frac{n!}{(n-2)! \cdot 2} = 6 \Rightarrow \frac{\overset{n(n-1)}{\cancel{n!}}}{\cancel{(n-2)!}} = 12 \Rightarrow$$

$\Rightarrow n(n-1) = 12$

$n^2 - n - 1 = 0$

$\Delta = 1 + 48 = 49$

$$n_{1,2} = \frac{1 \pm 7}{2} \begin{cases} 4 \\ -3 \end{cases}$$

$\Rightarrow n \in \mathbb{N} \Rightarrow n = 4$

a). $n = 4$

③. $(a_n)_{n \geq 1}$ $\in \frac{\cdot}{\circ}$ $r = 3$, $a_3 = 7$

$$a_n = (n-1) \cdot r + a_1$$

$$a_3 = 2 \cdot r + a_1 = 6 + a_1 \rightarrow a_1 = 1$$

$$a_5 = 4 \cdot r + a_1 = 4 \cdot 3 + 1 = 13$$

a). $a_5 = 13$

④. $\begin{vmatrix} x & 6 \\ 1 & 2 \end{vmatrix} = 0 \rightarrow 2x - 6 = 0 \rightarrow x = 3$

a). $x = 3$

⑤. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x + x^2$ $f'(1) = ?$

$$f'(x) = e^x + 2x$$

a). $e + 2$

$$f'(1) = e^1 + 2 \cdot 1 = e + 2$$

⑥. $l = \lim\limits_{x \to 2} \dfrac{x^2 - 3x + 2}{x - 2} \overset{\frac{0}{0}}{\underset{L'H}{=}} \lim\limits_{x \to 2} \dfrac{2x - 3}{1} = 1$

a) $l = 1$

7. $f:(1,\infty)\to\mathbb{R}$ $\quad f(x)=x+\dfrac{1}{x-1}$

Aria intre G_f și asimptotă oblică a sa.

$$m=\lim_{x\to\infty}\frac{x+\frac{1}{x-1}}{x}=\lim_{x\to\infty}1+\frac{\frac{1}{x-1}}{x}=\lim_{x\to\infty}1+\frac{1}{\underbrace{x(x-1)}_{\to 0}}=1$$

$$n=\lim_{k\to\infty}\cancel{x}+\frac{1}{x-1}-\cancel{x}=\lim_{x\to\infty}\frac{1}{x-1}=0$$

$y=x \to$ asimptotă oblică

Avem: $\displaystyle\int_2^3 x+\frac{1}{x-1}-x\,dx=\int_2^3 dx=\ln(x-1)\Big|_2^3$

$=\ln(2)-\ln(1)=\ln 2$

a). $\ln 2$

(8). x_1, x_2, x_3 – rădăcinile polinomului f, unde:

$f = x^3 + mx + 6$, $m < 0$

$x_1^4 + x_2^4 + x_3^4 = 98$ $\quad |x_1| + |x_2| + |x_3| = ?$

$$\sum_{i=1}^{3} x_i^4 = 98$$

$\forall i \in \{1, 2, 3\}$ avem $\quad x_i^3 + m \cdot x_i + 6 = 0 \mid \cdot x_i$

$$x_i^4 + m x_i^2 + 6 x_i = 0$$

$$x_i^4 = -m x_i^2 - 6 x_i$$

$$\sum_{i=1}^{3} x_i^4 = -\left(m \cdot \sum_{i=1}^{3} x_i^2 + 6 \cdot \sum_{i=1}^{3} x_i\right)$$

$$= -m(S_1^2 - 2S_2) - 6 \cdot S_1$$

unde $\xrightarrow{\text{Viette}}$ $\begin{cases} S_1 = x_1 + x_2 + x_3 = 0 \\ S_2 = x_1 x_2 + x_2 x_3 + x_3 x_1 = +m \\ S_3 = x_1 x_2 x_3 = -6 \end{cases}$

$$\Rightarrow x_1^4 + x_2^4 + x_3^4 = -m(S_1^2 - 2S_2) - 6 \cdot \underbrace{S_1}_{=0} =$$

$= \cancel{2}m^2 = 98 \mid :2$

$m^2 = 49 \xRightarrow{m<0} m = -7$

$\Rightarrow f = x^3 - 7 + 6$ Obs! că $x_1 = 4$ prima rădăcină

$$
\begin{array}{l|l}
x^3 - 7x + 6 & x - 1 \\
\cline{2-2}
-x^3 + x^2 & x^2 + x - 6 \\
\cline{1-1}
x^2 - 7x + 6 & \\
-x^2 + x & \\
\cline{1-1}
-6x + 6 & \\
+6x - 6 & \\
\cline{1-1}
= \quad = &
\end{array}
$$

$f = (x-1)(x^2 + x - 6)$

$x^2 + x - 6 = 0$

$\Delta = 1 + 24 = 25$

$x_{1,2} = \dfrac{-1 \pm 5}{2} = \begin{cases} 2 \\ -3 \end{cases}$

$\Rightarrow \begin{array}{l} x^1 = 1 \\ x^2 = 2 \\ x^3 = -3 \end{array}$

$\Rightarrow |x_1| + |x_2| + |x_3| = |1| + |2| + |-3| = 6$

a). 6

9. $\left[\frac{2x-1}{3}\right] + \left[\frac{4x+1}{6}\right] = \frac{5x-4}{3}$; nr. sol. = ?

Vrem să ducem ecuația la o formă de genul:

$[x] + [x + \frac{1}{2}]$ pentru a folosi identitatea lui

Hermite : $[x] + [x + \frac{1}{2}] = [2x]$

La model general :

$[x] + [x + \frac{1}{2}] + [x + \frac{2}{n}] + \dots [x + \frac{n-1}{2}] = [nx]$

Revenim: $\left[\frac{2x-1}{3}\right] + \left[\frac{4x+1}{6}\right] =$

$= \left[\frac{2x-1}{3}\right] + \left[\frac{4x+1}{6} - \frac{1}{2} + \frac{1}{2}\right] =$

$= \left[\frac{2x-1}{3}\right] + \left[\frac{4x+1-3}{6} + \frac{1}{2}\right] =$

$= \left[\frac{2x-1}{3}\right] + \left[\frac{4x-2}{6} + \frac{1}{2}\right] =$

$= \left[\frac{2x-1}{3}\right] + \left[\frac{2x-1}{3} + \frac{1}{2}\right] = \left[\frac{2(2x-1)}{3}\right]$

Avem: $\underbrace{\left[\frac{4x-2}{3}\right]}_{\in \mathbb{Z}} = \underbrace{\frac{5x-4}{3}}_{\in \mathbb{Z}}$

Ştim: $[a] \leq a < [a]+1$

$$\Rightarrow \left[\frac{4x-2}{3}\right] \leq \frac{4x-2}{3} < \left[\frac{4x-2}{3}\right] \mid \cdot 3$$

$$\parallel$$

$$\frac{5x-4}{3} \leq \frac{4x-2}{3} < \frac{5x-4+3}{3} \mid \cdot 3$$

$$5x-4 \leq 4x-2 < 5x-1 \mid -5x$$

$$-4 \leq -x-2 < -1 \mid \cdot -1$$

$$4 \geq x+2 > 1 \mid -2$$

$$2 \geq x > -1$$

$$\frac{5x-4}{3} \in \mathbb{Z} \Rightarrow \frac{5x-4}{3} \overset{k \in \mathbb{Z}}{=} k$$

$$\frac{5x-4}{3} = k \mid \cdot 3$$

$$5x-4 = 3k$$

$$x = \frac{3k+4}{5}, \text{ unde } x \in [-1, 2]$$

$$\left.\begin{array}{l} \dfrac{3k+4}{5} = -1 \Rightarrow 3k = -9 \Rightarrow k = -3 \\[2ex] \dfrac{3k+4}{5} = 2 \Rightarrow k = 2 \end{array}\right\} \Longrightarrow$$

$\Longrightarrow k \in \{-2, -1, 0, 1, 2\}$ avem 5 soluții

$x > -1 \Rightarrow k > -3$

$$x \in \left\{-\frac{2}{5}; \frac{1}{5}; \frac{4}{5}; \frac{7}{5}; 2\right\} \Rightarrow$$

$\Rightarrow$ 5 soluții a)

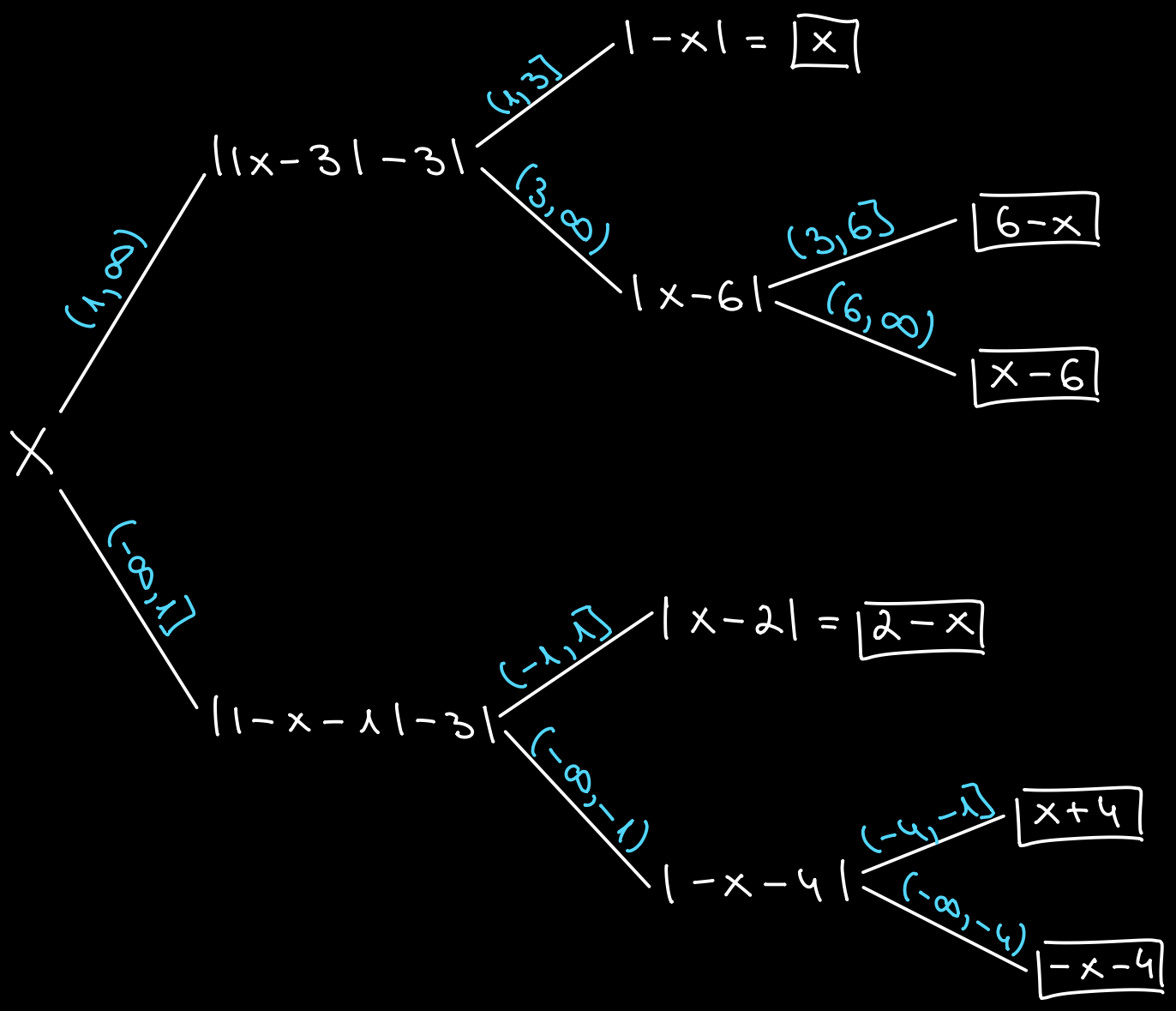
10. $||||x-1|-2|-3| = mx$

$m = ?$ a.î. avem o infinitate de soluții

notăm $f(x) = g(x) - mx$,

unde $g(x) = ||||x-1|-2|-3|$

Explicitare modul:



$$g(x) = \begin{cases} x-6, & x \in [6,\infty) \\ 6-x, & x \in [3,6) \\ x, & x \in [1,3) \\ 2-x, & x \in [-1,1) \\ x+4, & x \in [-4,-1) \\ -x-4, & x \in (-\infty,-4) \end{cases}$$

Dar $f(x) = g(x) - mx \Rightarrow$

$$\Rightarrow f(x) = \begin{cases} x-6-mx, & x \in [6,\infty) \\ 6-x-mx, & x \in [3,6) \\ x-mx, & x \in [1,3) \\ 2-x-mx, & x \in [-1,1) \\ x+4-mx, & x \in [-4,-1) \\ -x-4-mx, & x \in (-\infty,-4) \end{cases}$$

Obs! că $\forall\, [1,3)$, $f(x) = x - mx$

Dacă luăm $m=1$ vom avea: $f(x) = x - x = 0 \quad \forall\, x \in [1,3)$

Deci vom avea o inf. de soluții $\forall\, x \in [1,3)$ și $m=1$

a) $m=1$